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INTRUSION DETECTION SYSTEM

Network Malicious Detection (Benign vs Malicious)

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All IDSAI Dataset shape: (1000000, 26)

Dataset after drop duplicates: (693116, 26)

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Label (Benign and Malicious) counts:

label

1 471256

0 221860

Name: count, dtype: int64

Features after cleaning: 21

Data split: Train=485181 (70%), Test=207935 (30%)

Training model...

Training time: 132.98 seconds

Cross-Validation (on training set)...

10-Fold CV Accuracy on Train: 0.9994 ± 0.0001

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PERFORMANCE SUMMARY

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Accuracy | Train: 99.95% | Test: 99.95%

Balanced Acc | Train: 99.94% | Test: 99.94%

Precision | Train: 99.95% | Test: 99.95%

Recall | Train: 99.95% | Test: 99.95%

F1 | Train: 99.95% | Test: 99.95%

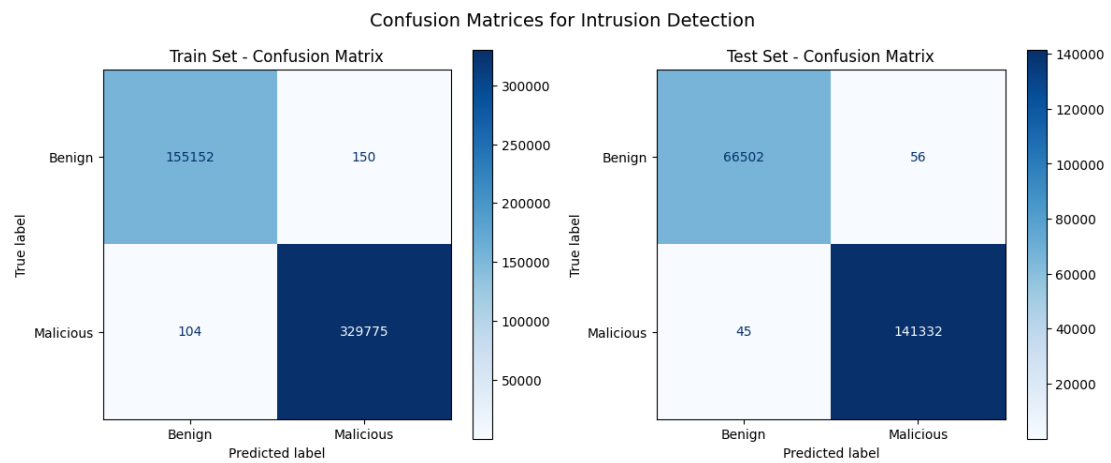
MCC | Train: 99.88% | Test: 99.89%

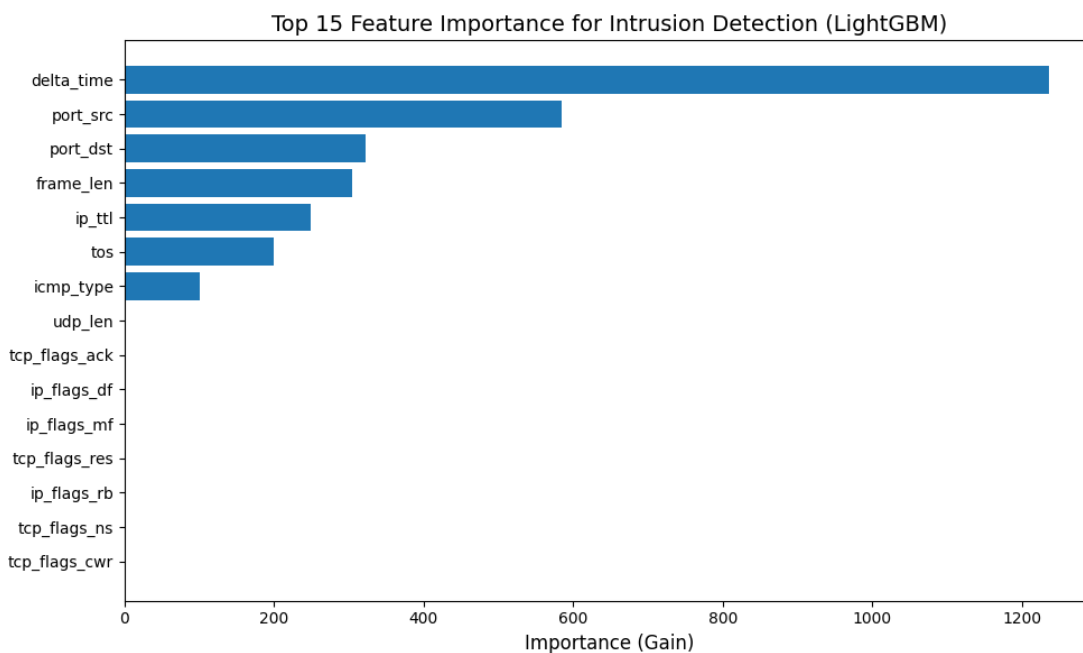
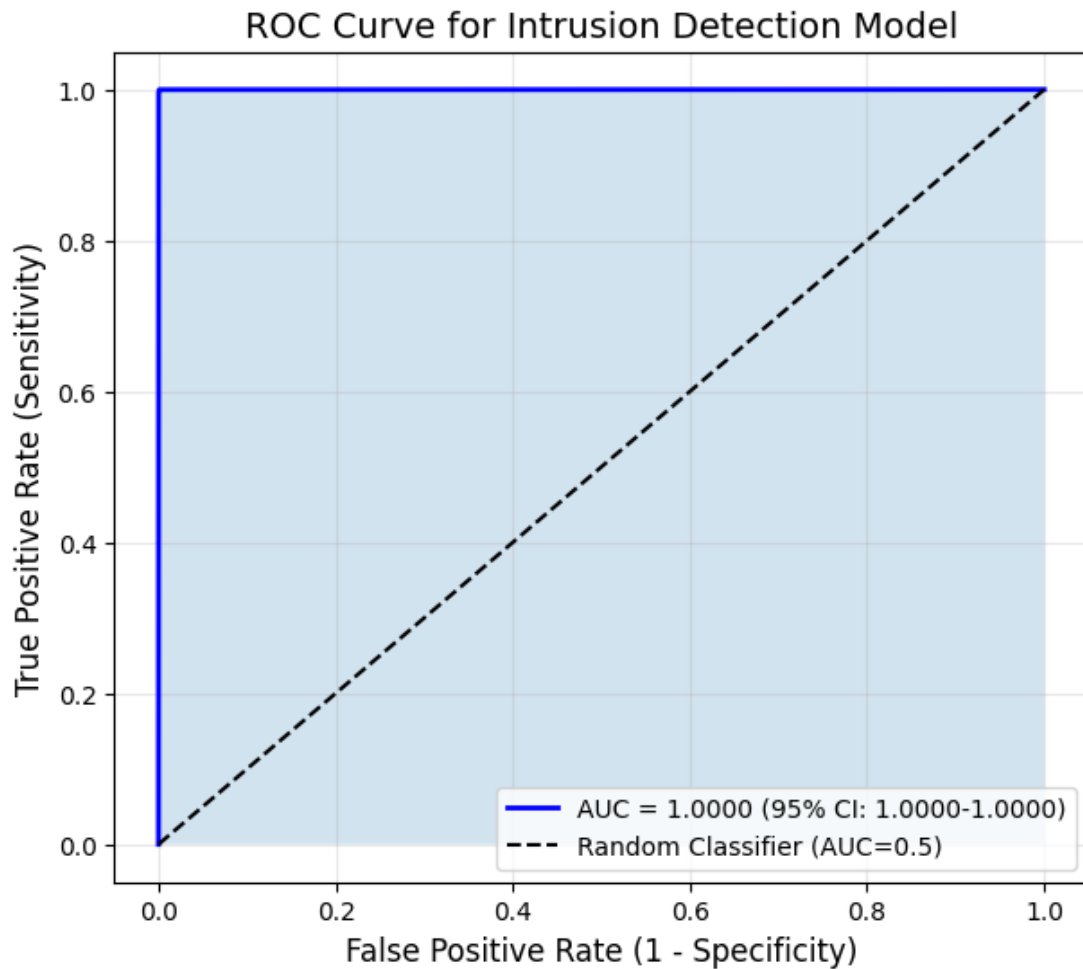
Kappa | Train: 99.88% | Test: 99.89%

Specificity | Train: 99.90% | Test: 99.92%

Sensitivity | Train: 99.97% | Test: 99.97%

AUC | Train: 0.00% | Test: 100.00%





Top 10 features:

Feature Importance

0	delta_time	1236
1	port_src	585
2	port_dst	322
3	frame_len	304
5	ip_ttl	249
7	tos	200
6	icmp_type	100
4	udp_len	2
16	tcp_flags_ack	2
9	ip_flags_df	0

SHAP analysis (sampling 100 test samples)...

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FEATURE IMPORTANCE BASED ON SHAP

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Feature	Strength	Impact %
frame_len	3.2581	47.41
port_dst	1.3421	19.53
tos	0.9327	13.57
port_src	0.7628	11.10
ip_ttl	0.3880	5.65
icmp_type	0.1389	2.02
delta_time	0.0479	0.70
tcp_flags_ack	0.0019	0.03
udp_len	0.0000	0.00

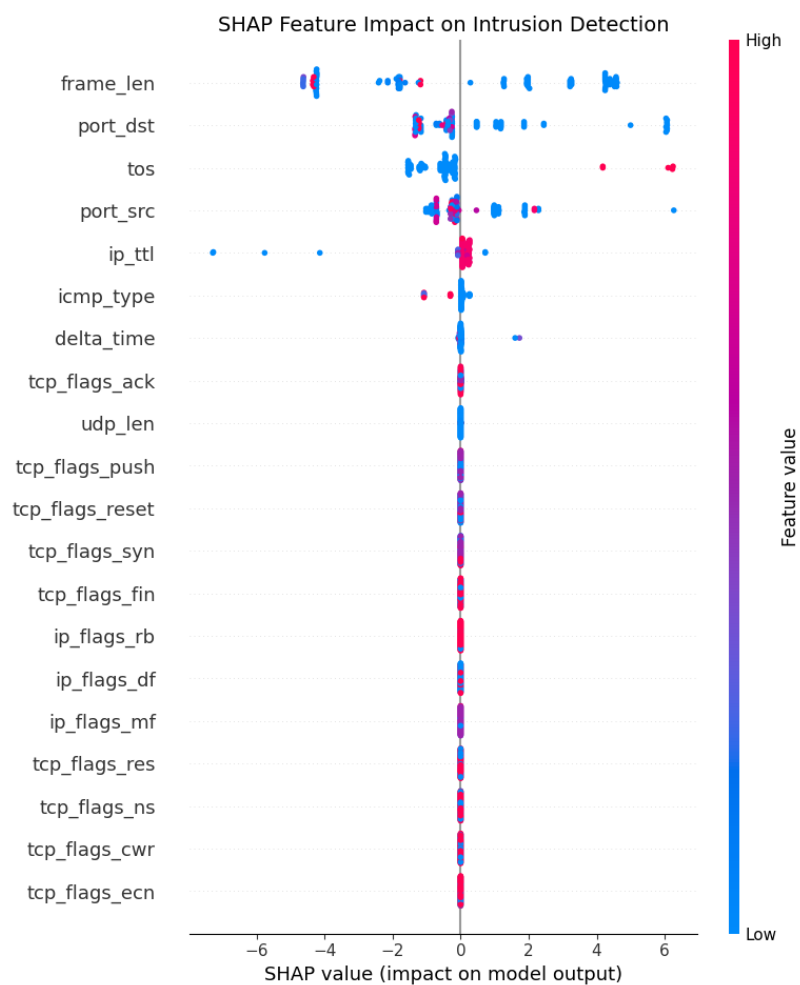
ip_flags_df	0.0000	0.00
ip_flags_mf	0.0000	0.00
tcp_flags_res	0.0000	0.00
ip_flags_rb	0.0000	0.00
tcp_flags_ns	0.0000	0.00
tcp_flags_cwr	0.0000	0.00

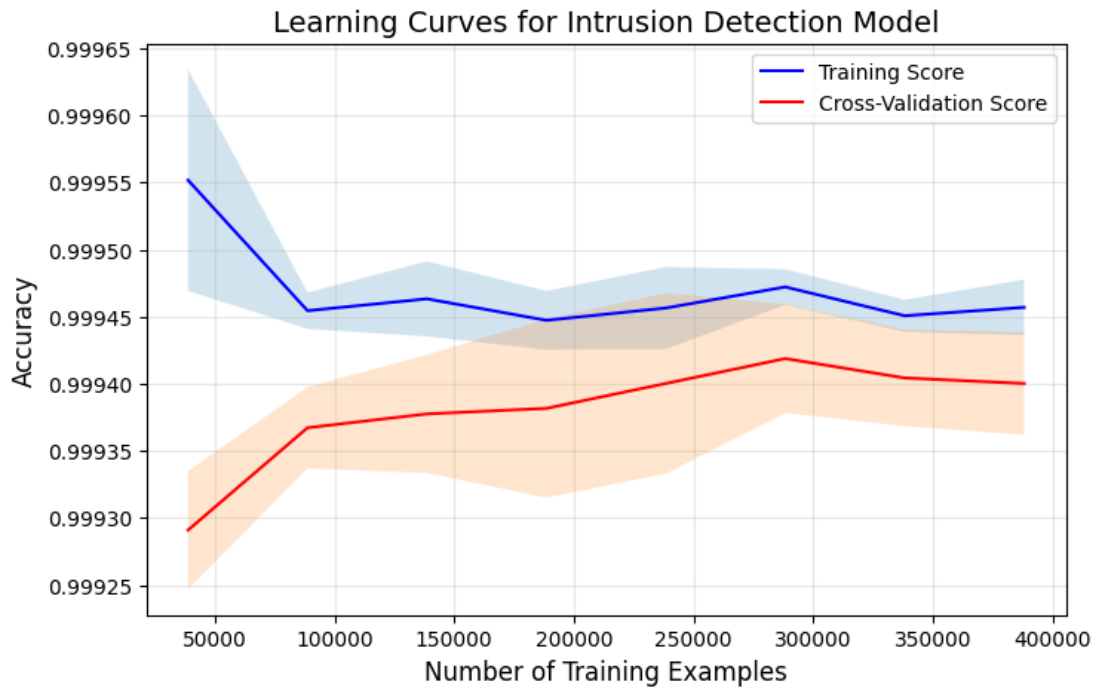
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Top feature: frame_len (Impact: 47.41%)

Second feature: port_dst (Impact: 19.53%)

Third feature: tos (Impact: 13.57%)





Cross-Validation Stability Comparison...

Cross Validation Results

	CV Std
xgb	0.000089
lgb	0.000103
Ensemble	0.000076

McNemar's test (Ensemble vs Logistic Regression): p-value = 0.000000

Result: Statistically significant difference ($p < 0.05$)

Model saved to fast_ids_model.pkl

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FINAL CLASSIFICATION REPORT (Test Set)

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precision	recall	f1-score	support
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Benign	0.98	0.99	0.98	527
Malicious	1.00	1.00	1.00	291345
accuracy			1.00	291872
macro avg	0.99	1.00	0.99	291872
weighted avg	1.00	1.00	1.00	291872

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INTRUSION DETECTION SYSTEM - COMPLETED SUCCESSFULLY

Results saved in: ids_results/

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Sample predictions on test instances:

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Total execution time: 11.45 minutes

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McNemar's test (Ensemble vs Logistic Regression): p-value = 0.000000

Result: Statistically significant difference ($p < 0.05$)

Model saved to fast_ids_model.pkl

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FINAL CLASSIFICATION REPORT (Test Set)

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	precision	recall	f1-score	support
Benign	1.00	1.00	1.00	66558
Malicious	1.00	1.00	1.00	141377
accuracy			1.00	207935
macro avg	1.00	1.00	1.00	207935
weighted avg	1.00	1.00	1.00	207935

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INTRUSION DETECTION SYSTEM - COMPLETED SUCCESSFULLY

Results saved in: ids_results/

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Sample predictions on test instances:

Correct | True: Benign | Predicted: Benign
Correct | True: Malicious | Predicted: Malicious
Correct | True: Malicious | Predicted: Malicious
Correct | True: Benign | Predicted: Benign
Correct | True: Benign | Predicted: Benign

Total execution time: 14.17 minutes

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